ST5202 —Applied Regression Analysis
Lecture Notes 9: Generalized Linear Model

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Outline

Motivation and framework

Binary response and logistic regression

Count response and Poisson regression

Why do we need generalized linear models?

- In multiple linear regression, we assume that the response variable is approximately *normal*, with mean

$$E(Y \mid \boldsymbol{x}) = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p.$$

- This framework is often inappropriate when Y is:
  - binary, such as success/failure, yes/no, disease/no disease;
  - a count, such as number of injuries, insurance claims, or accidents.
- the normality assumption and the constant variance assumption are usually unrealistic.
- We need a model that keeps the linear predictor, but allows a non-normal response distribution and a suitable mean structure.

Two common situations

- **Binary response:** the response takes only two values, usually coded as 0 and 1.
- Example: whether an individual gets a given disease or not
- **Count response:** the response is a non-negative integer.
- Example: the number of insurance claims in one year
- These two types of data are the main focus of this lecture.

- Other possible responses: waiting time of a customer, etc.

The basic idea of a GLM

- A generalized linear model (GLM) has three components:
  1. a response distribution;
  2. a linear predictor

$$\eta = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p;$$

  3. a link function relating the mean to the linear predictor:

$$g(\mu) = \eta, \qquad \mu = E(Y \mid \boldsymbol{x}).$$

- Thus, GLM extends linear regression by replacing "normal + identity link" with a more suitable combination.

Common GLMs in practice

- **Linear regression:** normal distribution with identity link.

$$g(\mu) = \mu$$

- **Logistic regression:** Bernoulli/binomial distribution with logit link.

$$g(\mu) = \log \frac{\mu}{1-\mu}$$

- **Poisson regression:** Poisson distribution with log link.

$$g(\mu) =$$

- These models all use the same linear predictor, but differ in the response distribution and the link function.

Binary response data

- For binary data, the response takes values 0 and 1.
- Let
$$Y_i \sim \text{Bernoulli}(\mu_i), \qquad P(Y_i = 1) = \mu_i,$$
where $0 < \mu_i < 1$.
- The mean and variance are
$$E(Y_i) = \mu_i, \qquad \text{Var}(Y_i) = \mu_i(1 - \mu_i).$$
- The variance depends on the mean, so the constant variance assumption of linear regression does not hold.

Why not use a linear probability model?

- One could try the model

$$\mu_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip}.$$

- This is called a linear probability model.
- It has two main problems:
  - fitted values may fall outside the interval $[0, 1]$;
  - the variance $\mu_i(1 - \mu_i)$ is not constant.
- Hence a linear model is usually not appropriate for binary response data.

Logistic regression model

- The most common GLM for binary data is logistic regression.
- The model is

$$Y_i \sim \text{Bernoulli}(\mu_i), \qquad \log \frac{\mu_i}{1-\mu_i} = \eta_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip}.$$

- The link function

$$g(\mu) = \log \frac{\mu}{1-\mu}$$

  is called the **logit link**.
- Correspondingly, the estimated probability follows that

$$\mu_i = \frac{\exp(\eta_i)}{1+\exp(\eta_i)}.$$

  . It is always between 0 and 1.

R procedure for binary data

- If the response is in ungrouped form, it is represented by a vector of 0's and 1's.
- The basic syntax is

```
glm(y ~ model.formula, family = binomial(logit),
    data = dataframe)
```

- Other links are also available for the binomial family:

  `logit, probit, cauchit, cloglog`

  They indicate different functions $g(\mu)$.

Example: space shuttle Challenger

- *The Challenger accident*. The accident occurred on January 28, 1986. At 11:39 EST about 73 seconds after launch, the space shuttle exploded and crashed into the Atlantic Ocean off the coast of Florida, killing all seven astronauts aboard.
- The cause of the accident was eventually traced to the failure of O-rings on its solid rocket booster. The O-rings failed because they lost flexibility at low temperatures, and the temperature that morning was 31Â°F, far below the lowest temperature recorded for previous launches.
- The temperature at 24 launches or static tests preceding the Challenger launch along with O-ring failure (0, no failure; 1 failure) were given below:

T(F^o)	53	56	57	63	66	67	67	67	68	69	70	70
Ind.	1	1	1	0	0	0	0	0	0	0	0	1
T(F^o)	70	70	72	73	75	75	76	76	78	79	80	81
Ind.	1	1	0	0	0	1	0	0	0	0	0	0

17. Challenger example: data

The data frame is created as follows:

```
ct=matrix(c(53,1,70,1,
            56,1,70,1,
            57,1,72,0,
            63,0,73,0,
            66,0,75,0,
            67,0,75,1,
            67,0,76,0,
            67,0,76,0,
            68,0,78,0,
            69,0,79,0,
            70,0,80,0,
            70,1,81,0), ncol=2,byrow=T)
Challenger=data.frame(temp = ct[,1],oring.f=ct[,2])
```

Challenger example: fitting the model in R

- We fit the logistic regression model

$$\log \frac{\mu_i}{1-\mu_i} = \beta_0 + \beta_1 \, temp_i.$$

- The R code is

```
EX1.fit=glm(oring.f ~ temp, family=binomial(logit),
            data=Challenger)
summary(EX1.fit)
vcov(EX1.fit)
```

- The estimated coefficients and standard errors are shown on the next slide.

Challenger example: fitted results

- Estimated coefficients:

```
Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept) 10.87535    5.70291   1.907   0.0565
temp        -0.17132    0.08344  -2.053   0.0400
```

- Estimated covariance matrix:

```
            (Intercept)         temp
(Intercept)  32.5232111 -0.473880521
temp         -0.4738805  0.006962009
```

Interpreting the fitted logistic model

- The fitted model is

$$\log \frac{\hat{\mu}(t)}{1-\hat{\mu}(t)} = 10.8754 - 0.1713\, t.$$

- Thus the fitted **log-odds** decreases as temperature increases.
- Equivalently, the fitted **odds** is

$$O(t) = \frac{\hat{\mu}(t)}{1-\hat{\mu}(t)} = \exp(10.8754 - 0.1713t).$$

- A higher launch temperature is associated with a lower chance of O-ring failure.

Odds and odds ratio

- For binary data, an important quantity is the **odds**

$$O(x) = \frac{\mu(x)}{1 - \mu(x)}.$$

- The ratio of two odds is called the **odds ratio**.
- In a logistic regression model, regression coefficients are naturally interpreted through odds ratios.

Interpreting a continuous predictor

- For the Challenger example,

$$\frac{O(t+1)}{O(t)} = \exp(-0.1713) = 0.8426.$$

- Thus, when temperature increases by $1^\circ F$, the odds of O-ring failure is multiplied by about 0.843.
- Equivalently, the odds decreases by about 15.7% for each one-degree increase in temperature.
- This is the standard interpretation of a slope coefficient in logistic regression.

Inference for the temperature effect

- From the R results, β_1 is significant at level $\alpha > 0.04$, that is, the temperature is significant for causing the O-ring failure.
- Confidence intervals for β_1 at level α:

$$[\hat{\beta}_1 - z_{\alpha/2} sd(\hat{\beta}_1),\ \hat{\beta}_1 + z_{\alpha/2} sd(\hat{\beta}_1)].$$

  $\hat{\beta}_1 = -0.1713$, $sd(\hat{\beta}_1) = 0.08344$. At $\alpha = 0.05$, $z_{\alpha/2} = 1.96$. The 95% confidence interval for β_1:

$$[-0.3348424,\ -0.0077576].$$

- Confidence interval for odds ratio:

$$[e^{\hat{\beta}_1 - z_{\alpha/2} sd(\hat{\beta}_1)},\ e^{\hat{\beta}_1 + z_{\alpha/2} sd(\hat{\beta}_1)}].$$

  The 95% confidence interval for the odds ratio:

$$[e^{-0.3348424},\ e^{-0.0077576}] = [0.7155,\ 0.9923].$$

Predicted probabilities in R

- The fitted probabilities can be extracted by

```
fitted(EX1.fit)
predict(EX1.fit, type="response")
```

- For a new temperature value, say 31 degrees Fahrenheit,

```
predict(EX1.fit,
        newdata=data.frame(temp=31),
        type="response")
```

- This gives the estimated probability of failure at launch.

Treatment of abdominal aneurysms (TAA): Data

- The data contain observations on 102 surgical control subjects (standard surgical procedure is used) and 266 subjects treated by endovascular approach.
- Response: whether or not the incidence of moderate to severe bleeding complications occurs within 30 days of surgery (`comp30` $= 1$ if moderate to severe, and 0 if not).
- Covariates:
  - Treatment group: (`control` $= 1$ if subject had standard procedure and 0 if endovascular approach),
  - Gender (`male` $= 1$ if subject is male and 0 if female),
  - Early investigator experience (`early` $= 1$ if investigator had limited experience and 0 if investigator had more experience),
  - Baseline aneurysm diameter (`diameter`).

TAA: Data at a glance

SEX	diameter	TREATMNT	comp30	male	control	early	conmale	conearly
M	50	Control	0	1	1	0	1	0
M	48.9	Control	0	1	1	0	1	0
M	59.2	Control	0	1	1	0	1	0
M	59	Control	0	1	1	0	1	0
M	48.1	Control	0	1	1	0	1	0
M	81	Control	1	1	1	0	1	0
M	55	Control	0	1	1	0	1	0
M	44	Control	1	1	1	0	1	0
M	57.7	Control	0	1	1	0	1	0
M	60.5	Control	0	1	1	0	1	0
M	45	Control	0	1	1	0	1	0
M	53.7	Control	0	1	1	0	1	0
M	54.6	Control	0	1	1	0	1	0
F	59	Control	0	0	1	0	0	0
M	46	Control	0	1	1	0	1	0
M	51.2	Control	1	1	1	0	1	0
M	53	Control	0	1	1	0	1	0
M	56.8	Control	1	1	1	0	1	0
F	43	Control	0	0	1	0	0	0
F	56.6	Control	1	0	1	0	0	0
M	58.7	Control	0	1	1	0	1	0
M	50	Control	1	1	1	0	1	0
M	53	Control	0	1	1	0	1	0
M	62.5	Control	0	1	1	0	1	0
M	48.8	Control	0	1	1	0	1	0
F	43.5	Control	0	0	1	0	0	0
M	55	Control	0	1	1	0	1	0
M	74	Control	0	1	1	0	1	0
M	47	Control	1	1	1	0	1	0

TAA example: R procedure

- Read the data into a data frame:

```
Aneurysm.dat=read.csv(file="D://Rsessions/Aneurysm.csv",
                      header = TRUE)
```

- There are 11 subjects with the values of `diameter` missing. Delete these subjects:

```
tt=na.omit(Aneurysm.dat)
```

- Fit the model:

```
EX2.fit=glm(comp30~control+male+early+diameter,
            family=binomial(logit),data=tt)
summary(EX2.fit)
round(vcov(EX2.fit),4)
```

TAA example: fitted results

- Coefficient estimation:

```
            Estimate Std. Error z value Pr(>|z|)
(Intercept) -2.60990    0.84137  -3.102 0.001922 **
control      1.69107    0.32189   5.254 1.49e-07 ***
male        -1.05385    0.34455  -3.059 0.002223 **
early        1.24298    0.35161   3.535 0.000408 ***
diameter     0.02418    0.01350   1.791 0.073230 .
```

- Covariance matrix:

```
            (Intercept) control    male   early diameter
(Intercept)      0.7079 -0.0683 -0.0854 -0.0670  -0.0102
control         -0.0683  0.1036  0.0001  0.0606   0.0001
male            -0.0854  0.0001  0.1187 -0.0109  -0.0001
early           -0.0670  0.0606 -0.0109  0.1236   0.0003
diameter        -0.0102  0.0001 -0.0001  0.0003   0.0002
```

TAA: Inference

- The fitted model:

$$\begin{aligned} \ln \frac{\pi}{1-\pi} &= -2.610 + 1.691 * \text{control} - 1.054 * \text{male} \\ &\quad +1.024 * \text{early} + 0.024 * \text{diameter}. \end{aligned}$$

- Surgery type (variable `control`), Gender (variable `male`), Surgeon's experience (variable `early`) are quite significant, especially the Surgery type.
- The effect of `diameter` does not seem to be statistically significant enough judging by the p-value 0.07323. However, as an adjustment variable, it should not be excluded from the model.

Adjusted odds ratios

- The odds ratio between two groups with different values of a given covariate while the values of the other covariates are fixed the same is called the adjusted odds ratio.
- The adjusted odds ratio of standard surgery and endovascular surgery:

$$\frac{e^{-2.610+1.691-1.054*m+1.024*e+0.024*d}}{e^{-2.610-1.054*m+1.024*e+0.024*d}} = e^{1.691} = 5.425.$$

- The 95% confidence interval for the odds ratio:

$$e^{[1.691 \pm 1.96*0.3219]} = [2.88659, 10.19527]$$

- In general, with logistic link, e^{β_j} is the adjusted odds ratio of the groups determined by the corresponding covariate.

Odds ratio vs adjusted odds ratio

- An **odds ratio** compares the odds of an event between two groups using only these two variables. It does **not** take other explanatory variables into account.
- An **adjusted odds ratio** is obtained from a multiple logistic regression model. It compares the odds between two groups **when the other variables in the model are kept the same**.
- If other explanatory variables are related to both the response and the predictor of interest, the odds ratio and the adjusted odds ratio can be quite different.

Adjusted odds ratios for the TAA example

- The main adjusted odds ratios are:

Effect	Estimate	95% C.I.	
Control	5.425	2.887	10.196
Male	0.349	0.177	0.685
Early	3.466	1.740	6.905

- Interpretation should always be made conditional on the other variables in the model.
- Take procedure as an example: The group with the standard procedure is more likely to have bleeding complications; when the other variables are kept the same, the odds of bleeding complications with the standard procedure are about 5.42 times those with the endovascular approach.

Logistic regression diagnostics

- The purpose of diagnostics is similar to that in linear regression:
  - check whether the model fits adequately;
  - identify unusual or influential observations;
  - assess whether important structure has been missed.
- The specific tools are different because the response is binary.

Remarks on logistic regression

- Logistic regression is the standard GLM for binary response data.
- The model is interpreted on the log-odds scale.
- Exponentiated coefficients give odds ratios or adjusted odds ratios.
- In practice, fitted probabilities, confidence intervals, and diagnostics should all be examined.

Count response data

- Characteristics of count data:
  - Count data are non-negative integers.
  - There is no obvious upper limit for the count data.
  - Usually, the mean of the data is proportional to its variance, which is the most important nature of count data.
- Examples:
  - Traffic accidents in a given period.
  - Number of insurance claims.
  - Number of customers visiting a shop.
- Data with the characteristics of counts can all be dealt with by the GLM with Poisson distributions

The Poisson distribution

- A Poisson random variable has probability mass function

$$P(Y = y) = e^{-\mu}\mu^y/y!, \qquad y = 0, 1, 2, \ldots$$

- Its mean and variance are both equal to μ:

$$E(Y) = \mu, \qquad \mathrm{Var}(Y) = \mu.$$

- Hence Poisson regression is suitable when the variance is roughly proportional to the mean.

Poisson regression model

- In Poisson regression,

$$Y_i \sim \text{Poisson}(\mu_i), \qquad \log(\mu_i) = \eta_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip}.$$

- The link function

$$g(\mu) = \log(\mu)$$

  is the log link.
- Under the log link,

$$\mu_i = \exp(\eta_i).$$

- Therefore the fitted mean is always positive.

R procedure for count data

- For Poisson regression, the family is `poisson`.
- The basic syntax is

```
glm(formula, family = poisson(log), data = dataframe)
```

- Other links are also available:

  `log, identity, sqrt`
- The log link is the most commonly used choice.

Mine fracture example

- Investigation was conducted to study how certain factors affect fractures in the upper seam of mines.
- The number of injuries or fractures that occur in the upper seam of mines in the coal fields of an area were observed. A total of 44 observations were collected on mines in this area. The following covariates were measured:

  x_1: Inner burden thickness in feet,
  x_2: Percent extraction of the lower previously mined seam,
  x_3: Lower seam height (feet),
  x_4: Time that the mine has been opened (years).
- A portion of the data is displayed on the next slide. The full Data is provided in the Excel file: `MineFracture.csv`

Mine fracture study: Data (partial)

y	x1	x2	x3	x4
2	50	70	52	1
1	230	65	42	6
0	125	70	45	1
4	75	65	68	0.5
1	70	65	53	0.5
2	65	70	46	3
0	65	60	62	1
0	350	60	54	0.5
4	350	90	54	0.5
4	160	80	38	0
1	145	65	38	10
4	145	85	38	0
1	180	70	42	2
5	43	80	40	0
2	42	85	51	12
5	42	85	51	0
5	45	85	42	0
5	83	85	48	10
0	300	65	68	10

Mine fracture: model selection

- Initial models:

```
library(MASS)
null.fit=glm(y~1,family=poisson(log),data=mine.dat)
full.fit=glm(y~.,family=poisson(log),data=mine.dat)
```

- Forward selection:

```
forward=stepAIC(null.fit, scope=list(upper=
    ~x1+x2+x3+x4, lower=~1), direction="forward")
summary(forward)
```

- Backward selection:

```
backward=stepAIC(full.fit, scope=list(upper=
    ~x1+x2+x3+x4, lower=~1),direction="backward")
summary(backward)
```

- Both procedure select the variables: `x1, x2, x4`.

34. Mine fracture: Fitting of the selected model

- The R codes:

```
mine.fit=glm(y~x1+x2+x4,family=poisson(log),
             data=mine.dat)
summary(mine.fit)
```

- The results:

```
              Estimate Std. Error z value Pr(>|z|)
(Intercept) -3.7206821  0.9788770  -3.801 0.000144
x1          -0.0014793  0.0008244  -1.794 0.072757
x2           0.0627011  0.0122711   5.110 3.23e-07
x4          -0.0316514  0.0163095  -1.941 0.052298
```

Mine fracture: Conclusion

- There are very strong evidence that the variable x_2, percent extraction of the lower previously mined seam, has an significant effect on the fracture of the upper seam. The p-value of the significant test for the effect is `3.23e-07`.
- There are also strong evidence for the significance of the effects of x_1, inner burden thickness, and x_4, time the mine has been opened. The p-values of the significant tests for the effects are, respectively, 0.072757 and 0.052298.
- Remark: when drawing conclusions, one should not stick to a given significance level such as 0.05 or 0.01 too strictly.

Interpreting coefficients in Poisson regression

- In a Poisson regression model, a coefficient is interpreted through a multiplicative change in the mean count.
- For example, if one predictor increases by one unit, while the other variables remain fixed,

$$\frac{\mu(x+1)}{\mu(x)} = e^{\beta}.$$

- Therefore $e^{\beta} - 1$ gives the proportional change in the mean count.

Interpreting the extraction effect in the mine example

- For variable x_2, the fitted coefficient is 0.0627.
- Hence
$$\frac{\mu(x_2+1)}{\mu(x_2)} = e^{0.0627} = 1.0647.$$
- Thus, for each one-percent increase in extraction, the mean number of fractures increases by about 6.47%, with the other variables fixed.
- This interpretation is often more natural than interpreting the coefficient itself.

Interpreting negative coefficients

- For x_1 and x_4, the fitted coefficients are negative.
- This means that increasing these variables decreases the mean number of fractures.
- For example, an increase of 10 units in x_4 changes the mean by a factor
$$e^{10(-0.03165)} \approx 0.729.$$
- So the mean number of fractures is reduced by about 27.1%.

Rates, exposures, and offsets

- Sometimes the observed count depends on an exposure level, such as time at risk, population size, or area.
- Then it is often more meaningful to model a **rate** rather than a raw count.
- If exposure is m_i, we write

$$\log(\mu_i) = \log(m_i) + \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip}.$$

- The term $\log(m_i)$ is called an offset. The coefficient of the offset is fixed at 1.
- In R, an offset is included by

```
glm(y ~ x1 + x2 + x3 + offset(log(exposure)),
    family=poisson(log), data=dataframe)
```